

MF RECOGNITION IS Π_2^0 -COMPLETE

SAUERS

A countable group is MF if it admits finite-dimensional unitary models that are asymptotically multiplicative and asymptotically separate nonidentity elements in operator norm. The finite presentations of MF groups form a Π_2^0 -complete set, and their complement is Σ_2^0 -complete. MF recognition from finite presentations is undecidable. Neither the MF presentations nor the non-MF presentations can be recursively enumerated. For each n , a finite certificate gives normal-closure proofs for words declared trivial and requires a unitary model that separates every remaining word, with relator error at most 2^{-n} . Validity is decidable by Tarski's theorem. We also construct, computably from any program, a finite presentation that defines an MF group if and only if the program halts on infinitely many inputs. If the program halts on only finitely many inputs, the presented group contains the non-MF group of [NMF]; if it halts on infinitely many inputs, the group is MF by a permanence theorem for HNN extensions whose edge isomorphism is implemented by a unitary of a norm matrix corona.

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1. INTRODUCTION

Not every countable group is MF: *Non-MF Groups* [NMF] constructs a finitely generated simple group $H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2))$ with property (T) such that every homomorphism from H to an MF group is trivial. We turn to the recognition problem for finitely presented groups.

Theorem A (MF recognition is Π_2^0 -complete). *Deciding whether a finite presentation defines an MF group is Π_2^0 -complete, and the complementary problem is Σ_2^0 -complete. MF recognition from finite presentations is undecidable. Neither the MF presentations nor the non-MF presentations can be recursively enumerated. There is a computable map $e \mapsto \hat{R}_e$ such that $\hat{R}_e$ presents an MF group if and only if the e -th partial computable function has infinite domain.*

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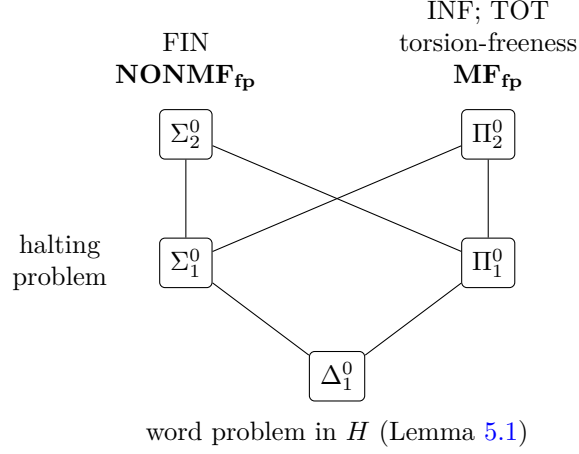


FIGURE 1. The arithmetical hierarchy. Lines indicate inclusions, and each listed problem is complete at its level; the word problem in H is decidable. The two sets from Theorem A appear in bold. $\text{MF}_{\text{fp}} \in \Pi_2^0$ by finite certificates (Section 3). The family $\widehat{R}_e$ reduces INF to MF_{fp} and FIN to NONMF_{fp} (Section 5).

For each n , a certificate consists of a dimension d , a label for every word of length at most n , and a normal-closure proof for every word labelled trivial. It is valid if there are unitaries $U_1, \dots, U_k \in \mathcal{U}(d)$ such that $\|r_i(U) - 1\| \leq 2^{-n}$ for each relator r_i and $\|w(U) - 1\| \geq 1/4$ for every word labelled nontrivial. These conditions are polynomial equations and inequalities over the real numbers, so validity is decidable by Tarski's theorem. A finite presentation defines an MF group if and only if it has a valid certificate for every n . Hence $\text{MF}_{\text{fp}} \in \Pi_2^0$ and $\text{NONMF}_{\text{fp}} \in \Sigma_2^0$ (Section 3).

Given a program index e , we construct a finite presentation $\widehat{R}_e$ in Section 5. If the e -th partial function has finite domain, then the presented group contains H and is not MF. If its domain is infinite, the edge isomorphism in the defining HNN extension is implemented by a unitary of a norm matrix corona, and the presented group is MF by Theorem 4.2. Korchagin proved that the MF property is closed under direct limits [Kor, Proposition 13]. Consequently, every finitely generated non-MF group is a quotient of a finitely presented one.

Relation to prior work. Dehn initiated the study of decision problems for finitely presented groups. Adian and Rabin later proved that every Markov property is undecidable from a finite presentation [Ad, Ra]. A property is Markov if it holds for some finitely presented group and fails for every group containing some fixed finitely presented group. The trivial group is MF, whereas every group containing H is non-MF. Thus MF is a Markov

property, and the Adian–Rabin theorem proves undecidability. Our theorem determines the precise level of undecidability.

Lempp proved that recognizing torsion-free finitely presented groups is Π_2^0 -complete [Lem]. For recursively presented groups, Bilanovic–Chubb–Roven proved that every Markov property is Π_2^0 -hard to recognize [BCR]; their construction begins our reduction. We must then replace a recursive presentation by a finite one without changing whether the group is MF. A general Higman embedding does not preserve the MF property. We therefore use HNN extensions chosen to satisfy the permanence theorem proved below.

2. MF GROUPS

A countable group G is *MF* if there are positive integers d_n and maps $V_n: G \rightarrow \mathcal{U}(d_n)$, with $V_n(1) = 1$, such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \rightarrow 0 \quad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \quad (g \in G \setminus \{1\}).$$

This is the operator norm notion of MF group introduced by Carrión–Dadarlat–Eckhardt [CDE]. Here, MF group always means this operator-norm local-approximation property. The strong-convergence convention also requires the matrix models to recover the norms of the left regular representation [GKM, Sch]. Equivalently, G embeds in the unitary group of the norm matrix corona associated to the sequence $\mathbf{d} = (d_n)$,

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}),$$

where $\bigoplus_n M_{d_n}(\mathbb{C})$ is the c_0 -direct sum of the sequences with $\|x_n\| \rightarrow 0$. A separable C^* -algebra is called MF if it embeds as a C^* -subalgebra of a norm matrix corona [BK]. Restricting models to a subgroup shows that subgroups of MF groups are MF.

Throughout, the binary Leavitt algebra $L_{\mathbb{F}_2}(1, 2)$ is the unital $\mathbb{F}_2$ -algebra generated by s_0, s_1, t_0, t_1 subject to $t_i s_j = \delta_{ij}$ and $s_0 t_0 + s_1 t_1 = 1$ [Lev], and $H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2))$ is the subgroup of $\text{GL}_{12}(L_{\mathbb{F}_2}(1, 2))$ generated by the elementary matrices $e_{ij}(a) = 1 + aE_{ij}$, $i \neq j$. By [NMF, Theorem B], every homomorphism from H to an MF group is trivial; in particular H is not MF.

3. LOCALITY AND FINITE CERTIFICATES

We use the following finite-presentation form of Korchagin's local criterion [Kor, Propositions 7–8].

Lemma 3.1 (Korchagin's criterion). *Let $G = \langle x_1, \dots, x_k \mid r_1, \dots, r_m \rangle$. Then G is MF if and only if the following holds. Given $\varepsilon > 0$ and finitely many words $w_1, \dots, w_q$ representing nonidentity elements of G , there are $d \geq 1$ and $U_1, \dots, U_k \in \mathcal{U}(d)$ such that*

$$\|r_i(U) - 1\| \leq \varepsilon \quad (1 \leq i \leq m), \quad \|w_j(U) - 1\| \geq \frac{1}{2} \quad (1 \leq j \leq q).$$

Proof. Korchagin proves the criterion with one chosen word representing each element of a finite subset of G , and with separation constant 1 [Kor, Propositions 7–8]. If ω_j is his chosen representative of the element represented by w_j , then $w_j \omega_j^{-1}$ is a finite product of conjugates of the relators and their inverses. For the finite family under consideration, taking the relator tolerance sufficiently small therefore preserves separation at least $1/2$ for the prescribed words. Conversely, Korchagin's amplification argument replaces any fixed positive separation constant by 1 while changing the relator tolerance only by a constant factor. $\square$

A set $A \subseteq \mathbb{N}$ belongs to Π_2^0 if there is a decidable relation R such that

$$x \in A \iff \forall n \exists c R(x, n, c).$$

Its complement then belongs to Σ_2^0 ; see [So]. Fix a computable coding of finite presentations by natural numbers. From a code P we can recover the generators and relators of the presentation. Write G_P for the presented group and set

$$\text{MF}_{\text{fp}} = \{P : G_P \text{ is MF}\}, \quad \text{NONMF}_{\text{fp}} = \mathbb{N} \setminus \text{MF}_{\text{fp}}.$$

Proposition 3.2 (finite certificates). *There is a decidable predicate $C(P, n, c)$ such that*

$$P \in \text{MF}_{\text{fp}} \iff \forall n \exists c C(P, n, c).$$

Consequently $\text{MF}_{\text{fp}} \in \Pi_2^0$ and $\text{NONMF}_{\text{fp}} \in \Sigma_2^0$.

Proof. Fix $P = \langle x_1, \dots, x_k \mid r_1, \dots, r_m \rangle$, and let W_n be the finite set of reduced words of length at most n . A certificate contains the following data:

- (1) a positive integer d ;
- (2) a partition $W_n = T \sqcup S$;
- (3) for each $w \in T$, an expression of w as a product of conjugates of the relators and their inverses.

The certificate is valid if every expression in item 3 freely reduces to the indicated word and there are $U_1, \dots, U_k \in \mathcal{U}(d)$ satisfying

$$\|r_i(U) - 1\| \leq 2^{-n} \quad (1 \leq i \leq m), \quad \|w(U) - 1\| \geq \frac{1}{4} \quad (w \in S).$$

Here $w(U)$ denotes evaluation at the matrices U_j , with x_j^{-1} evaluated as U_j^* .

Free reduction decides whether the expressions in item 3 are correct. Every word admitted to T is therefore trivial in G_P . It remains to decide whether the required matrices exist. Write $U_j = A_j + iB_j$ with real matrices A_j and B_j . Unitarity becomes a finite system of polynomial equations. For $t \geq 0$, the norm bounds can be written as

$$\begin{aligned} \|X\| \leq t &\iff t^2 I - X^* X \succeq 0, \\ \|X\| \geq t &\iff \exists v (\|v\| = 1 \text{ and } \|Xv\|^2 \geq t^2). \end{aligned}$$

Positive semidefiniteness is determined by the principal minors. After splitting real and imaginary parts, all these conditions are polynomial equalities and inequalities over $\mathbb{R}$. The existence of the matrices is therefore a sentence in the first-order theory of real closed fields, which is decidable by Tarski's theorem [Ta]. Let $C(P, n, c)$ mean that c encodes a valid certificate. Then C is decidable.

Suppose first that G_P is MF. Put every trivial word in W_n into T and every remaining word into S . Each word in T has a normal-closure expression of the required form. Applying Lemma 3.1 to the finite family S , with relator tolerance 2^{-n} , supplies the matrices required for a valid certificate.

Conversely, suppose that a valid certificate exists for every n . Let $w_1, \dots, w_q$ represent nonidentity elements of G_P , and let $\varepsilon > 0$. Choose n larger than the lengths of these words and such that $2^{-n} \leq \varepsilon$. None of the w_j can lie in T , since membership in T includes a proof that the word is trivial. Thus every w_j lies in S , and the matrices in the certificate satisfy the relator and separation bounds in Lemma 3.1, with separation $1/4$ in place of $1/2$. Korchagin's amplification argument allows any fixed positive separation constant, so G_P is MF.

We have proved the displayed equivalence. It places MF_{fp} in Π_2^0 and its complement in Σ_2^0 . $\square$

4. HNN EXTENSIONS WITH A CORONA CONJUGATOR

In [NMF] we prove non-approximability using one-sided conjugation in a norm matrix corona. If the edge isomorphism of an HNN extension is implemented by a unitary of a norm matrix corona, then the HNN extension is MF. For $\theta = \text{id}$ this is close to Shulman's theorem on central HNN extensions [Sh]; the version below assumes a tracial state vanishing off the identity and allows a nontrivial edge isomorphism. The corona unitary implements the edge isomorphism, while the trace and the left regular representation prove that the group embeds in the resulting algebra.

A countable group G admits a tracial MF realization (A, ρ, τ) if A is a separable unital C^* -algebra, $\rho: G \rightarrow \mathcal{U}(A)$ is a homomorphism, and τ is a tracial state on A with $\tau(\rho(g)) = 0$ for every $g \neq 1$. Since $\tau(1) = 1$, such a group embeds in $\mathcal{U}(A)$, so it is MF. Only the vanishing of τ off the identity is used below; traciality is what the constructions below provide. To embed an arbitrary norm matrix corona $\mathcal{Q}_{\mathbf{d}}$ in $\prod_k M_k(\mathbb{C}) / \bigoplus_k M_k(\mathbb{C})$, choose a

strictly increasing sequence (m_n) with $m_n \geq d_n$, embed $M_{d_n}(\mathbb{C})$ as a corner of $M_{m_n}(\mathbb{C})$, and put zero in the unused coordinates. The resulting map on reduced products is an isometric $*$ -homomorphism, so the MF algebras of Section 2 are exactly the separable C^* -algebras that embed in the latter corona, as in [BK, Sh].

Lemma 4.1 (residually finite groups). *Every countable residually finite group admits a tracial MF realization.*

Proof. Choose finite-index normal subgroups $N_1 \geq N_2 \geq \dots$ with trivial intersection, and let λ_k be the left regular representation of G/N_k , of dimension d_k . In the corona $\mathcal{Q}_{\mathbf{d}}$ the map $\rho(g) = [(\lambda_k(g))_k]$ is a homomorphism, and $\tau([(x_k)_k]) = \lim_{\omega} \text{tr}_{d_k}(x_k)$, for a free ultrafilter ω , is a tracial state with $\tau(\rho(g)) = 0$ for $g \neq 1$, since $g \notin N_k$ for all large k . The C^* -algebra generated by $\rho(G)$ and the unit of the corona is separable and unital, and the triple with ρ and τ is the required realization. $\square$

Let G be a countable group, let $S \leq G$ be a subgroup, and let $\theta: S \rightarrow G$ be an injective homomorphism. The HNN extension

$$(1) \quad R = \langle G, t \mid tst^{-1} = \theta(s) \ (s \in S) \rangle$$

contains G by Britton's lemma [Bri].

Theorem 4.2 (HNN permanence with a corona conjugator). *In the situation of (1), suppose that G admits a tracial MF realization (A, ρ, τ) , and that for some norm matrix corona $\mathcal{Q}$ there are an injective $*$ -homomorphism $\iota: A \rightarrow \mathcal{Q}$ and a unitary $W \in \mathcal{Q}$ with*

$$W \iota \rho(s) W^* = \iota \rho(\theta(s)) \quad (s \in S).$$

Then R admits a tracial MF realization. In particular, R is MF.

Proof. Write $D = C^*(\iota \rho(G)) \subseteq \mathcal{Q}$, a separable C^* -algebra with unit $p = \iota(1)$, and let $B_0 = C^*(\iota \rho(S))$ and $B_1 = C^*(\iota \rho(\theta(S)))$, both containing p . Conjugation by W restricts to a $*$ -isomorphism $\Theta: B_0 \rightarrow B_1$ carrying $\iota \rho(s)$ to $\iota \rho(\theta(s))$. Let U be the quotient of the full unital free product $D * C(\mathbb{T})$ by the closed ideal generated by the elements $ubu^* - \Theta(b)$, $b \in B_0$, where u is the canonical unitary generator of $C(\mathbb{T})$. This is the universal C^* -HNN extension of D associated to Θ ; we use d and u for the canonical images of $d \in D$ and the stable unitary. Let $C = B_0 \oplus B_1$, $A_1 = M_2(D)$, and $A_2 = M_2(B_0)$, with the unital inclusions

$$\iota_A(c_0, c_1) = \begin{pmatrix} c_0 & 0 \\ 0 & c_1 \end{pmatrix} \in A_1, \quad \iota_B(c_0, c_1) = \begin{pmatrix} c_0 & 0 \\ 0 & \Theta^{-1}(c_1) \end{pmatrix} \in A_2,$$

and let $P = A_1 *_C A_2$ be the full amalgamated free product, in which A_1 and A_2 embed [Sh, Section 2.1]. Ueda proved that the universal HNN extension is a full corner of exactly this amalgamated free product [Ue, Proposition 2.4]: for the projection $e = \iota_A(1_D, 0) = \iota_B(1_D, 0)$ there is an isomorphism of U onto ePe carrying $d \in D$ to $\text{diag}(d, 0)$ and the unitary u to the partial

isometry $e_{12}f_{21}$ built from the matrix units e_{ij} of A_1 and f_{ij} of A_2 . In particular U embeds in P .

Shulman's criterion reduces the MF property of P to compatible embeddings of A_1 and A_2 in one norm matrix corona [Sh, Theorem 16]. Identify $M_2(\mathcal{Q})$ with the norm matrix corona for the doubled dimension sequence. Let φ_A be the entrywise inclusion of $M_2(D)$, and let φ_B be the entrywise inclusion of $M_2(B_0)$ followed by conjugation by $\text{diag}(1, W)$. Both maps are injective, and for $(c_0, c_1) \in C$,

$$\varphi_B(\iota_B(c_0, c_1)) = \begin{pmatrix} c_0 & 0 \\ 0 & W\Theta^{-1}(c_1)W^* \end{pmatrix} = \begin{pmatrix} c_0 & 0 \\ 0 & c_1 \end{pmatrix} = \varphi_A(\iota_A(c_0, c_1)),$$

because Θ is conjugation by W . So P is MF, as is its C^* -subalgebra $ePe \cong U$.

The functional $T = \tau \circ \iota^{-1}$ is a tracial state on $\iota(A) \supseteq D$ with $T(\iota\rho(g)) = 0$ for $g \neq 1$. In the GNS representation of D for T , the vectors $\xi_g = \iota\rho(g)\xi_T$, $g \in G$, are orthonormal, because $\langle \xi_g, \xi_h \rangle = T(\iota\rho(h^{-1}g))$. Their closed span is invariant under D , and on it $\iota\rho(g)$ acts as the left regular representation $\lambda_G(g)$ of G . Thus there is a $*$ -homomorphism $\pi: D \rightarrow C_r^*(G)$ with $\pi(\iota\rho(g)) = \lambda_G(g)$. Since $G \leq R$, the space $\ell^2(R)$ is a direct sum of copies of $\ell^2(G)$ indexed by the right cosets of G , so $\lambda_R|_G$ is a multiple of λ_G and the assignment $\lambda_G(g) \mapsto \lambda_R(g)$ extends to a $*$ -homomorphism $C_r^*(G) \rightarrow C_r^*(R)$. Composing, we obtain $\sigma_0: D \rightarrow C_r^*(R)$ with $\sigma_0(\iota\rho(g)) = \lambda_R(g)$. The pair $(\sigma_0, \lambda_R(t))$ satisfies the covariance relation $\lambda_R(t)\sigma_0(b)\lambda_R(t)^* = \sigma_0(\Theta(b))$ for $b \in B_0$: on the generators $\iota\rho(s)$ both sides equal $\lambda_R(\theta(s))$, because $tst^{-1} = \theta(s)$ in R , and two $*$ -homomorphisms that agree on generators agree on B_0 . The universal property of U gives $\sigma: U \rightarrow C_r^*(R)$ with $\sigma(\iota\rho(g)) = \lambda_R(g)$ and $\sigma(u) = \lambda_R(t)$.

Define $j: R \rightarrow \mathcal{U}(U)$ by $j(g) = \iota\rho(g)$ for $g \in G$ and $j(t) = u$. The defining relations of (1) hold in U , since $u\iota\rho(s)u^* = \Theta(\iota\rho(s)) = \iota\rho(\theta(s))$, so j is a homomorphism, and $\sigma \circ j = \lambda_R$ is injective, so j is injective. Let $A' = C^*(j(R)) \subseteq U$, a separable unital MF C^* -algebra, and let $\tau' = \tau_R \circ \sigma|_{A'}$, where τ_R is the canonical trace $x \mapsto \langle x\delta_1, \delta_1 \rangle$ of $C_r^*(R)$. Then τ' is a tracial state on A' with $\tau'(j(r)) = \tau_R(\lambda_R(r)) = 0$ for $r \neq 1$. So (A', j, τ') is a tracial MF realization of R . $\square$

Injectivity of j follows from $\sigma \circ j = \lambda_R$, not from the covariant pair $(\iota\rho, W)$, which need not be faithful; in the applications below it is not.

Corollary 4.3 (central HNN extensions). *If G admits a tracial MF realization and $S \leq G$ is any subgroup, then $\langle G, t \mid [t, s] = 1 \ (s \in S) \rangle$ admits a tracial MF realization.*

Proof. Apply Theorem 4.2 with θ the inclusion of S , with ι any injective $*$ -homomorphism of A into a norm matrix corona, and with $W = 1$. $\square$

The positive branch of the reduction below produces tracial MF realizations inside norm reduced products. For C^* -algebras B_n , $n \geq 1$, write

$\prod_n B_n / \bigoplus_n B_n$ for the quotient of the bounded sequences by the sequences with $\|x_n\| \rightarrow 0$; its norm is $\limsup_n \|x_n\|$.

Lemma 4.4 (reduced products of MF algebras). *If every B_n is MF, then every separable C^* -subalgebra of $\prod_n B_n / \bigoplus_n B_n$ is MF. Moreover, $M_k(B)$ is MF whenever B is.*

The first assertion follows by diagonalization from the matrix-corona characterization of MF algebras (Blackadar–Kirchberg [BK]; see also [BO, Chapter 11]), and Korchagin uses it in this form [Kor, Proposition 6]. The second holds because M_k of a norm matrix corona is the norm matrix corona of the dimension sequence multiplied by k .

Lemma 4.5 (tensor synchronization). *Let Γ be a countable group with a tracial MF realization (A_1, ρ_1, τ_1) , let Q be a countable group with homomorphisms $\beta_n: Q \rightarrow B_n$ to finite groups such that every $q \neq 1$ satisfies $\beta_n(q) \neq 1$ for all large n , let $S \leq \Gamma$, and let $\tau: S \rightarrow Q$ be a homomorphism. Suppose that for every n there is a homomorphism $\lambda_n: \Gamma \rightarrow G_n$ to a finite group with*

$$\ker(\lambda_n|_S) \leq \ker(\beta_n \circ \tau).$$

Then there are a separable unital MF C^ -algebra A , a homomorphism $V: \Gamma \times Q \rightarrow \mathcal{U}(A)$, a tracial state T on A with $T(V(g, q)) = 0$ for $(g, q) \neq (1, 1)$, and a unitary $W \in A$ with $WV(s, 1)W^* = V(s, \tau(s))$ for all $s \in S$. Consequently*

$$\langle \Gamma \times Q, u \mid u(s, 1)u^{-1} = (s, \tau(s)) \ (s \in S) \rangle$$

admits a tracial MF realization, and in particular is MF.

Proof. Let E_n be the image of $(\lambda_n, \beta_n): \Gamma \times Q \rightarrow G_n \times B_n$, a finite group of order k_n , and let L_n be its left regular representation on $\ell^2(E_n)$. Put $B'_n = A_1 \otimes M_{k_n}(\mathbb{C})$, which is MF by Lemma 4.4, let $E = \prod_n B'_n / \bigoplus_n B'_n$, and define

$$V(g, q) = [(\rho_1(g) \otimes L_n(\lambda_n(g), \beta_n(q)))_n], \quad T([(x_n)_n]) = \lim_{\omega} (\tau_1 \otimes \text{tr}_{k_n})(x_n),$$

for a free ultrafilter ω . Each coordinate of V is a homomorphism into a unitary group, so V is a homomorphism, and T is a tracial state on E because $|(\tau_1 \otimes \text{tr}_{k_n})(x_n)| \leq \|x_n\|$. If $g \neq 1$ then $\tau_1(\rho_1(g)) = 0$, so $T(V(g, q)) = 0$; if $g = 1$ and $q \neq 1$ then $\beta_n(q) \neq 1$ for all large n , the regular representation of a finite group has trace zero off the identity, and again $T(V(1, q)) = 0$.

The two homomorphisms $s \mapsto (\lambda_n(s), 1)$ and $s \mapsto (\lambda_n(s), \beta_n \tau(s))$ from S to E_n have the same kernel, namely $\ker(\lambda_n|_S)$, by the hypothesis. So their images are isomorphic subgroups of E_n of the same order, isomorphic via $(\lambda_n(s), 1) \mapsto (\lambda_n(s), \beta_n \tau(s))$. The restriction of L_n to a subgroup of E_n is unitarily equivalent to the direct sum of as many copies of the left regular representation of that subgroup as it has right cosets, so there is a unitary $W_n \in M_{k_n}(\mathbb{C})$ with $W_n L_n(\lambda_n(s), 1) W_n^* = L_n(\lambda_n(s), \beta_n \tau(s))$ for all $s \in S$. Put $W = [(1 \otimes W_n)_n] \in E$; then $WV(s, 1)W^* = V(s, \tau(s))$ for $s \in S$. Let A be the C^* -algebra generated by $V(\Gamma \times Q)$ and W ; it is separable and unital, and MF by Lemma 4.4. The triple $(A, V, T|_A)$ is a tracial MF realization

of $\Gamma \times Q$. For the last assertion, apply Theorem 4.2 to this realization. Let ι be an injective $*$ -homomorphism of A into a norm matrix corona and put $p = \iota(1)$. The element $\widehat{W} = \iota(W) + (1 - p)$ is a unitary of the full corona and implements the required covariance on $\iota(A)$, so it is the required conjugator. $\square$

5. THE COMPLEXITY OF RECOGNIZING MF GROUPS

Write W_e for the domain of the e -th partial computable function and

$$\text{INF} = \{e : W_e \text{ is infinite}\}, \quad \text{FIN} = \{e : W_e \text{ is finite}\}.$$

The set INF is Π_2^0 -complete and FIN is Σ_2^0 -complete [So, Chapter IV]. The construction combines the non-MF group H of [NMF] with Theorem 4.2: it produces a computable map $e \mapsto \widehat{R}_e$ into finite presentations such that the presented group contains H when $e \in \text{FIN}$ and satisfies the hypotheses of Theorem 4.2 when $e \in \text{INF}$, so that it is MF if and only if $e \in \text{INF}$. For $e \in \text{FIN}$, the group $\widehat{R}_e$ is not MF; for $e \in \text{INF}$, Lemma 5.8 proves that it is MF. Table 1 lists the objects of the construction.

H	the non-MF group $\text{EL}_{12}(L_{\mathbb{F}_2}(1, 2))$ of [NMF]
C_e	recursively presented; trivial for $e \in \text{INF}$, contains H for $e \in \text{FIN}$
$Q_e = F/N_e$	three-generated recursively presented group containing C_e ; here $F = F(x, y, t)$
$Q_+ = F/N_+$	the value of Q_e for trivial C_e ; embeds in $F(x_1, y) \times F(x_2, t)$
$\Lambda_e; M_e$	finitely presented Higman host of Q_e ; the Mihailova subgroup detecting equality in Λ_e
$K_e; L_e$	a product of six free groups; its finitely generated subgroup with $i(F) \cap L_e = i(N_e)$
$\Gamma_e; S_e$	the central HNN extension $\langle K_e, v \mid [v, \ell] = 1 \ (\ell \in L_e) \rangle$; $S_e = \langle i(F), v i(F) v^{-1} \rangle \cong F *_{N_e} F$
$\alpha_e: S_e \rightarrow Q_e$	q_e on $i(F)$, trivial on $v i(F) v^{-1}$
$\widehat{R}_e$	the finite presentation of $\langle \Gamma_e \times Q_e, u \mid u(s, 1)u^{-1} = (s, \alpha_e(s)) \ (s \in S_e) \rangle$

TABLE 1. The objects of the construction, in order of appearance.

A recursive presentation of H .

Lemma 5.1 (a recursive presentation of H). *The group $H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2))$ is finitely generated and has decidable word problem. In particular, it has a recursive presentation on finitely many generators.*

Proof. Let Y be the set of elements $e_{ij}(a)$ with $i \neq j$ and $a \in \{1, s_0, s_1, t_0, t_1\}$. Since $e_{ij}(a + b) = e_{ij}(a)e_{ij}(b)$ and $[e_{ik}(a), e_{kj}(b)] = e_{ij}(ab)$ for distinct i, j, k , induction on monomial length shows that $H = \langle Y \rangle$. The monomials in

s_0, s_1, t_0, t_1 that contain no subword $t_i s_j$ and no subword $s_1 t_1$ form the standard reduced basis of $L_{\mathbb{F}_2}(1, 2)$ by the basis theorem for Leavitt path algebras, with s_1 chosen as the special edge [LN, Theorem 3.7]. The reductions $t_i s_j \rightarrow \delta_{ij}$ and $s_1 t_1 \rightarrow 1 + s_0 t_0$ compute products in this basis, so equality in $L_{\mathbb{F}_2}(1, 2)$ is decidable. Thus a word in $Y^{\pm 1}$ evaluates to an explicit 12×12 matrix over $L_{\mathbb{F}_2}(1, 2)$, and it represents 1 in H exactly when that matrix is the identity. The set of words representing 1 is decidable, so recursively enumerable, and $\langle Y \mid \text{that set} \rangle$ is a recursive presentation of H . $\square$

The group C_e .

Lemma 5.2 (the group C_e). *There is an algorithm that computes from e a recursive presentation, on a computable set of generators, of a group C_e such that C_e is trivial if $e \in \text{INF}$ and H embeds in C_e if $e \in \text{FIN}$.*

Proof. Apply the construction of Bilanovic–Chubb–Roven [BCR, Theorem 3.1] to the trivial group and H . Let $\langle Y \mid T \rangle$ be the recursive presentation of H from Lemma 5.1. Take generators $y_{i,\ell}$ for $i \geq 1$ and $\ell \in Y$, one copy Y_i of Y for each i , and as relators all words of T rewritten in each copy Y_i , together with every generator $y_{i,\ell}$ for which $i \leq |W_e|$. The last family is recursively enumerable uniformly in e : enumerate W_e , and whenever its i -th element appears, enumerate all of Y_i . The group presented is the free product of the copies $H_i \cong H$, $i \geq 1$, with the first $|W_e|$ copies killed. If W_e is infinite every copy is killed and C_e is trivial; if W_e is finite, C_e is the free product of the remaining copies and contains H . $\square$

A three-generator embedding. Let $F = F(x, y, t)$ be the free group on x, y, t , and let $y_i = x^i y x^{-i}$ for $i \in \mathbb{Z}$. For a countable group C with a generating sequence $(c_i)_{i \geq 1}$, put $c_i = 1$ for $i \leq 0$ and

$$(2) \quad B(C) = \langle C * F(x, y), t \mid t y_i t^{-1} = c_i y_i \ (i \in \mathbb{Z}) \rangle.$$

Let $Q_+ = B(1) = \langle x, y, t \mid [t, y_i] = 1 \ (i \in \mathbb{Z}) \rangle$, let $q_+ : F \rightarrow Q_+$ be the quotient map, and let $N_+ = \ker q_+$.

Lemma 5.3 (the three-generator embedding). (1) $B(C)$ is an HNN extension of $C * F(x, y)$, so C embeds in $B(C)$; it is generated by x, y, t ; and a recursive presentation of $B(C)$ on x, y, t is computable from a recursive presentation of C . We write $B(C) = F/N_C$ and $q_C : F \rightarrow B(C)$ for the quotient map.

- (2) Killing C induces a surjection $B(C) \rightarrow Q_+$; so $N_C \leq N_+$, with equality when C is trivial.
- (3) The assignment $x \mapsto (x_1, x_2)$, $y \mapsto (y, 1)$, $t \mapsto (1, t)$ defines an injective homomorphism $j : Q_+ \rightarrow P = F(x_1, y) \times F(x_2, t)$. In particular, Q_+ is residually finite.

Proof. (1) The elements y_i , $i \in \mathbb{Z}$, freely generate the normal closure of y in $F(x, y)$ [LS, Chapter I]. The retraction $C * F(x, y) \rightarrow F(x, y)$ killing C sends $c_i y_i$ to y_i , so a nontrivial reduced word in the $c_i y_i$ maps to a nontrivial word

in the y_i ; so the $c_i y_i$ freely generate a free subgroup, and $y_i \mapsto c_i y_i$ is an isomorphism between two free subgroups of the base. Thus (2) is an HNN extension, and the base, and with it C , embeds by Britton's lemma. The relation $c_i = t y_i t^{-1} y_i^{-1}$ shows that x, y, t generate $B(C)$, and substituting this expression for c_i into the relators of C and into (2) produces a recursive presentation on x, y, t , uniformly in the presentation of C .

(2) The relators of (2) map to the relators $[t, y_i]$ of Q_+ when every c_i is sent to 1, and the map is the identity on x, y, t .

(3) The relators $[t, y_i]$ are sent to $[(1, t), (x_1^i y x_1^{-i}, 1)] = 1$, so j is a homomorphism. Let N_0 be the normal closure of $\{y, t\}$ in Q_+ . Then $Q_+/N_0 = \langle x \rangle$ is infinite cyclic and the extension splits, so $Q_+ = N_0 \rtimes \langle x \rangle$. Rewriting the presentation of Q_+ over the transversal $\{x^n\}$ by the Reidemeister–Schreier method [LS, Chapter II] presents N_0 on the generators $y_n = x^n y x^{-n}$ and $t_m = x^m t x^{-m}$, $n, m \in \mathbb{Z}$, with relators the conjugates $x^m [t, y_n] x^{-m} = [t_m, y_{n+m}]$, that is, all commutators $[t_m, y_n]$. So $N_0 = F(y_n : n \in \mathbb{Z}) \times F(t_m : m \in \mathbb{Z})$. Now $j(y_n) = (x_1^n y x_1^{-n}, 1)$ and $j(t_m) = (1, x_2^m t x_2^{-m})$, and the elements $x_1^n y x_1^{-n}$ freely generate the normal closure of y in $F(x_1, y)$, and likewise for t ; so j restricted to N_0 is an isomorphism onto the product of these two normal closures. Finally $j(x) = (x_1, x_2)$ has infinite order and $j(x)^k \notin j(N_0)$ for $k \neq 0$, because, for $h \in N_0$, the exponent sum of x_1 in $j(x^k h)$ is k . So j is injective on $N_0 \rtimes \langle x \rangle$. Free groups are residually finite and residual finiteness passes to direct products and subgroups, so Q_+ is residually finite. $\square$

For $e \in \mathbb{N}$ let $Q_e = B(C_e) = F/N_e$ with C_e the group of Lemma 5.2, and write $q_e: F \rightarrow Q_e$. By Lemmas 5.2 and 5.3, a recursive presentation of Q_e on x, y, t is computable from e ; $N_e \leq N_+$; if $e \in \text{INF}$ then $Q_e = Q_+$, $N_e = N_+$ and $q_e = q_+$; and if $e \in \text{FIN}$ then $H \leq C_e \leq Q_e$.

Detecting N_e inside a finitely presented group.

Lemma 5.4 (Higman embedding and the Mihailova subgroup). *There is an algorithm that computes from e a finite presentation $\langle X_e \mid R_e \rangle$ of a group Λ_e , words $w_x, w_y, w_t \in F(X_e)$, and a finite generating set of a subgroup $M_e \leq F(X_e) \times F(X_e)$, such that $x \mapsto w_x$, $y \mapsto w_y$, $t \mapsto w_t$ induces an embedding $Q_e \rightarrow \Lambda_e$, and $M_e = \{(u, v) : u = v \text{ in } \Lambda_e\}$. Consequently, with*

$$K_e^0 = F \times F(X_e) \times F(X_e), \quad i_e(f) = (f, w_f, 1), \quad L_e^0 = F \times M_e,$$

where w_f is the image of f under $x \mapsto w_x$, $y \mapsto w_y$, $t \mapsto w_t$, the map $i_e: F \rightarrow K_e^0$ is injective and $i_e(f) \in L_e^0$ if and only if $f \in N_e$.

Proof. Higman's embedding theorem [Hig] embeds every finitely generated recursively presented group in a finitely presented group, and its proof is effective; an explicit algorithm producing the finite presentation and the embedding words from a recursive presentation is given by Mikaelian [Mik]. Apply it to the presentation of Q_e . Let M_e be the subgroup of $F(X_e) \times F(X_e)$ generated by the pairs (z, z) , $z \in X_e$, and $(r, 1)$, $r \in R_e$; then $M_e = \{(u, v) :$

$u = v$ in Λ_e by Mihailova's lemma [Mih]. The first coordinate makes i_e injective, and $(f, w_f, 1) \in F \times M_e$ if and only if $w_f = 1$ in Λ_e , if and only if $q_e(f) = 1$ because the embedding is injective, if and only if $f \in N_e$. $\square$

The two HNN extensions. Let $P = F(x_1, y) \times F(x_2, t)$ and $j: Q_+ \rightarrow P$ be as in Lemma 5.3, and put

$$K^g = F \times P, \quad L^g = \{(f, jq_+(f)) : f \in F\},$$

and define

$$K_e = K_e^0 \times K^g, \quad L_e = L_e^0 \times L^g, \quad i: F \rightarrow K_e, \quad i(f) = (i_e(f), (f, 1)).$$

Let

$$(3) \quad \Gamma_e = \langle K_e, v \mid [v, \ell] = 1 \ (\ell \in L_e) \rangle, \quad S_e = \langle i(F), v i(F) v^{-1} \rangle \leq \Gamma_e.$$

Lemma 5.5 (the first HNN extension). (1) K_e is a direct product of free groups of finite rank, so finitely presented and residually finite; L_e is finitely generated; i is injective; and $i(F) \cap L_e = i(N_e)$.

(2) Γ_e is finitely presented, and a finite presentation is computable from e . The natural map from the amalgamated free product $F *_{N_e} F$, in which $n \in N_e$ in the first copy is identified with n in the second copy, onto S_e , sending the first copy to $i(F)$ and the second to $v i(F) v^{-1}$, is an isomorphism.

(3) There is a homomorphism $\alpha_e: S_e \rightarrow Q_e$ with $\alpha_e(i(f)) = q_e(f)$ and $\alpha_e(v i(f) v^{-1}) = 1$ for $f \in F$.

Proof. (1) $K_e = F \times F(X_e) \times F(X_e) \times F \times P$ is a product of six free groups of finite rank, so it is finitely presented, and residually finite because free groups are. The subgroup L_e is generated by the basis of the first factor, the generators of M_e from Lemma 5.4, and the elements $(a, jq_+(a))$ for $a \in \{x, y, t\}$, so it is finitely generated. Injectivity of i is clear from the first coordinate. An element $i(f)$ lies in L_e if and only if $i_e(f) \in L_e^0$ and $(f, 1) \in L^g$; the first holds if and only if $f \in N_e$ by Lemma 5.4, and the second if and only if $jq_+(f) = 1$, that is, $f \in N_+$. Since $N_e \leq N_+$, the intersection is $i(N_e)$.

(2) The presentation (3) is finite once $[v, \ell] = 1$ is imposed only for the finitely many generators ℓ of L_e , and it is computable from e because K_e , L_e and i are. It is an HNN extension of K_e with both edge groups equal to L_e and the identity as edge isomorphism. The map from $F *_{N_e} F$ to S_e is well defined because $i(n) \in L_e$ commutes with v for $n \in N_e$. A reduced word of the amalgamated free product alternates between elements of the two copies of F lying outside N_e ; its image is a word $i(f_0) v i(f_1) v^{-1} i(f_2) \dots$ in which every letter between consecutive occurrences of $v^{\pm 1}$ is some $i(f_k)$ with $f_k \notin N_e$, so $i(f_k) \notin L_e$ by (1). Such a word contains no pinch $v^{\pm 1} \ell v^{\mp 1}$ with $\ell \in L_e$, so it is nontrivial in Γ_e by Britton's lemma.

(3) Define α_e on $F *_{N_e} F$ by q_e on the first copy and by the trivial map on the second; both kill N_e , so the definition is consistent on the amalgamated subgroup, and (2) identifies $F *_{N_e} F$ with S_e . $\square$

The second HNN extension is

$$(4) \quad R_e = \langle \Gamma_e \times Q_e, u \mid u(s, 1)u^{-1} = (s, \alpha_e(s)) \ (s \in S_e) \rangle,$$

whose edge maps $s \mapsto (s, 1)$ and $s \mapsto (s, \alpha_e(s))$ are injective homomorphisms of S_e into $\Gamma_e \times Q_e$; so $\Gamma_e \times Q_e$ embeds in R_e by Britton's lemma. The presentation (4) lists the recursively enumerable relators of Q_e . Its finite replacement is

$$(5) \quad \begin{aligned} \hat{R}_e = \langle \Gamma_e \times F, u \mid & u(i(a), 1)u^{-1} = (i(a), a), \\ & u(v i(a) v^{-1}, 1)u^{-1} = (v i(a) v^{-1}, 1) \ (a \in \{x, y, t\}) \rangle, \end{aligned}$$

where $\Gamma_e \times F$ is equipped with the finite presentation of Lemma 5.5(2), the basis x, y, t of F , and the commutation relators between the two factors.

Lemma 5.6 (the finite presentation). *The groups $\hat{R}_e$ and R_e are isomorphic by the map that is the identity on Γ_e and on u and sends F onto Q_e by q_e . A code of the finite presentation (5) is computable from e .*

Proof. Both maps $f \mapsto u(i(f), 1)u^{-1}$ and $f \mapsto (i(f), f)$ are homomorphisms $F \rightarrow \hat{R}_e$, and they agree on the basis, so $u(i(f), 1)u^{-1} = (i(f), f)$ for all $f \in F$; likewise $u(v i(f) v^{-1}, 1)u^{-1} = (v i(f) v^{-1}, 1)$ for all f . If $n \in N_e$, then $i(n) \in L_e$ by Lemma 5.5(1), so $v i(n) v^{-1} = i(n)$ in Γ_e , and comparing the two relations gives $(i(n), n) = (i(n), 1)$, that is, $(1, n) = 1$ in $\hat{R}_e$. So the factor F of $\hat{R}_e$ factors through $Q_e = F/N_e$, and the relations of (5) become $u(s, 1)u^{-1} = (s, \alpha_e(s))$ for s in the generating set $\{i(a), v i(a) v^{-1} : a \in \{x, y, t\}\}$ of S_e ; both sides define homomorphisms on S_e , so the relations hold for all $s \in S_e$. This defines a homomorphism $R_e \rightarrow \hat{R}_e$; conversely, the relations of (5) hold in R_e after $F \rightarrow Q_e$, because $\alpha_e(i(a)) = q_e(a)$ and $\alpha_e(v i(a) v^{-1}) = 1$. The two homomorphisms are mutually inverse on generators. Computability of the code follows from Lemmas 5.4 and 5.5. $\square$

Lemma 5.7 (the case $e \in \text{FIN}$). *If $e \in \text{FIN}$, then H embeds in $\hat{R}_e$, and $\hat{R}_e$ is not MF.*

Proof. For $e \in \text{FIN}$ we have $H \leq C_e \leq Q_e \leq \Gamma_e \times Q_e \leq R_e \cong \hat{R}_e$ by Lemmas 5.2, 5.3(1), and 5.6, and Britton's lemma for (4). Restricting MF models to a subgroup shows that subgroups of MF groups are MF, and H is not MF by [NMF, Theorem B]. $\square$

Lemma 5.8 (the case $e \in \text{INF}$). *If $e \in \text{INF}$, then $\hat{R}_e$ admits a tracial MF realization; in particular, it is MF.*

Proof. Let $e \in \text{INF}$, so that $Q_e = Q_+$, $N_e = N_+$, $q_e = q_+$, and $\alpha_e(i(f)) = q_+(f)$, $\alpha_e(v i(f) v^{-1}) = 1$. By Lemma 5.6 it suffices to show that R_e admits a tracial MF realization, and by Lemma 4.5 applied to $\Gamma = \Gamma_e$, $Q = Q_+$, $S = S_e$ and $\tau = \alpha_e$, it suffices to produce the data of that lemma.

The group K_e is residually finite by Lemma 5.5(1), so it admits a tracial MF realization by Lemma 4.1; and Γ_e , the central HNN extension of K_e over L_e , admits one by Corollary 4.3.

The group $P = F(x_1, y) \times F(x_2, t)$ is residually finite; choose finite-index normal subgroups $P_1 \geq P_2 \geq \dots$ of P with trivial intersection, let $r_n: P \rightarrow C_n = P/P_n$, and put $\beta_n = r_n \circ j: Q_+ \rightarrow C_n$ with j the embedding of Lemma 5.3(3). Since j is injective and the P_n decrease to the trivial group, every $h \neq 1$ has $\beta_n(h) \neq 1$ for all large n .

Let $G_n = (C_n \times C_n) \rtimes \langle \sigma \rangle$, where σ of order two exchanges the coordinates. Define λ_n on $K_e = K_e^0 \times (F \times P)$ by killing K_e^0 and sending (f, p) to $(r_n j q_+(f), r_n(p))$, and put $\lambda_n(v) = \sigma$. This respects the relations of (3): the factor L_e^0 of L_e lies in K_e^0 and is killed, and an element $(f, j q_+(f))$ of L_e^g is sent to $(r_n j q_+(f), r_n j q_+(f))$, which lies on the diagonal and commutes with σ . So $\lambda_n: \Gamma_e \rightarrow G_n$ is a homomorphism, and on the generators of S_e ,

$$\lambda_n(i(f)) = (r_n j q_+(f), 1), \quad \lambda_n(v i(f) v^{-1}) = (1, r_n j q_+(f)).$$

Let $\pi_0, \pi_1: S_e \rightarrow Q_+$ be the homomorphisms that are q_+ on one copy of F in $S_e \cong F *_{N_+} F$ and trivial on the other, as in Lemma 5.5(3); thus $\pi_0 = \alpha_e$. The displayed formulas show that $\lambda_n|_{S_e} = (r_n j \pi_0, r_n j \pi_1)$, both sides being homomorphisms that agree on generators. So

$$\ker(\lambda_n|_{S_e}) = \ker(r_n j \pi_0) \cap \ker(r_n j \pi_1) \leq \ker(\beta_n \circ \alpha_e),$$

which is the hypothesis of Lemma 4.5. $\square$

Proof of Theorem A. By Lemma 5.6 the map $e \mapsto \hat{R}_e$ is computable, by Lemma 5.8 the presented group is MF for $e \in \text{INF}$, and by Lemma 5.7 it is not MF for $e \in \text{FIN}$. So INF reduces to MF_{fp} and FIN reduces to NONMF_{fp} under computable many-one reductions. Since INF is Π_2^0 -complete and FIN is Σ_2^0 -complete, and since $\text{MF}_{\text{fp}} \in \Pi_2^0$ and $\text{NONMF}_{\text{fp}} \in \Sigma_2^0$ by Proposition 3.2, the two sets are complete at their levels. A Π_2^0 -complete set is not Σ_2^0 and a Σ_2^0 -complete set is not Π_2^0 , so neither set is recursively enumerable, and in particular neither is decidable. $\square$

Remark 5.9. The construction uses only these properties of H : it is finitely generated, recursively presented, and not MF, while subgroups of MF groups are MF. So it applies to any group property that is closed under subgroups, given a finitely generated recursively presented group without the property and an analogue of Lemma 4.5 for the twisted HNN extensions; the open problems below record which input is missing for soficity and for hyperlinearity.

The homomorphism V of Lemma 4.5 does not extend faithfully to R_e in the corona: the finite maps λ_n are trivial on the Mihailova factor, so W commutes with $V(\gamma, 1)$ for every $\gamma \in K_e^0$, although u does not commute with $(\gamma, 1)$ in R_e . Injectivity is obtained in Theorem 4.2 from the universal algebra.

Open problems. We do not know whether the models of $\widehat{R}_e$ can be chosen to converge to $C_r^*(\widehat{R}_e)$ in operator norm, so that the reduced C^* -algebra itself is MF; the argument produces only a tracial state on an abstract MF algebra. We do not know whether $\widehat{R}_e$ is hyperlinear for $e \in \text{INF}$; placing hyperlinearity recognition at the same level would also require a finitely presented non-hyperlinear group, and none is known. We do not know whether an analogous family of finitely presented groups exists for soficity; such a construction would require a permutation-model replacement for Lemma 4.5. Finally, we do not know the complexity of MF recognition under the strong convergence convention of [GKM, Sch]: the norms $\|\lambda(w)\|$ in the left regular representation are not in general computable from a finite presentation, so Proposition 3.2 does not apply.

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USE OF AI AND FORMAL METHODS

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